|INCH-POUND|

MS24140L
9 June 1992
SUPERSEDING
MS24140K
20 February 1990

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 50 AMPERES, 1 PST (N.O.), TYPE I, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consisit of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

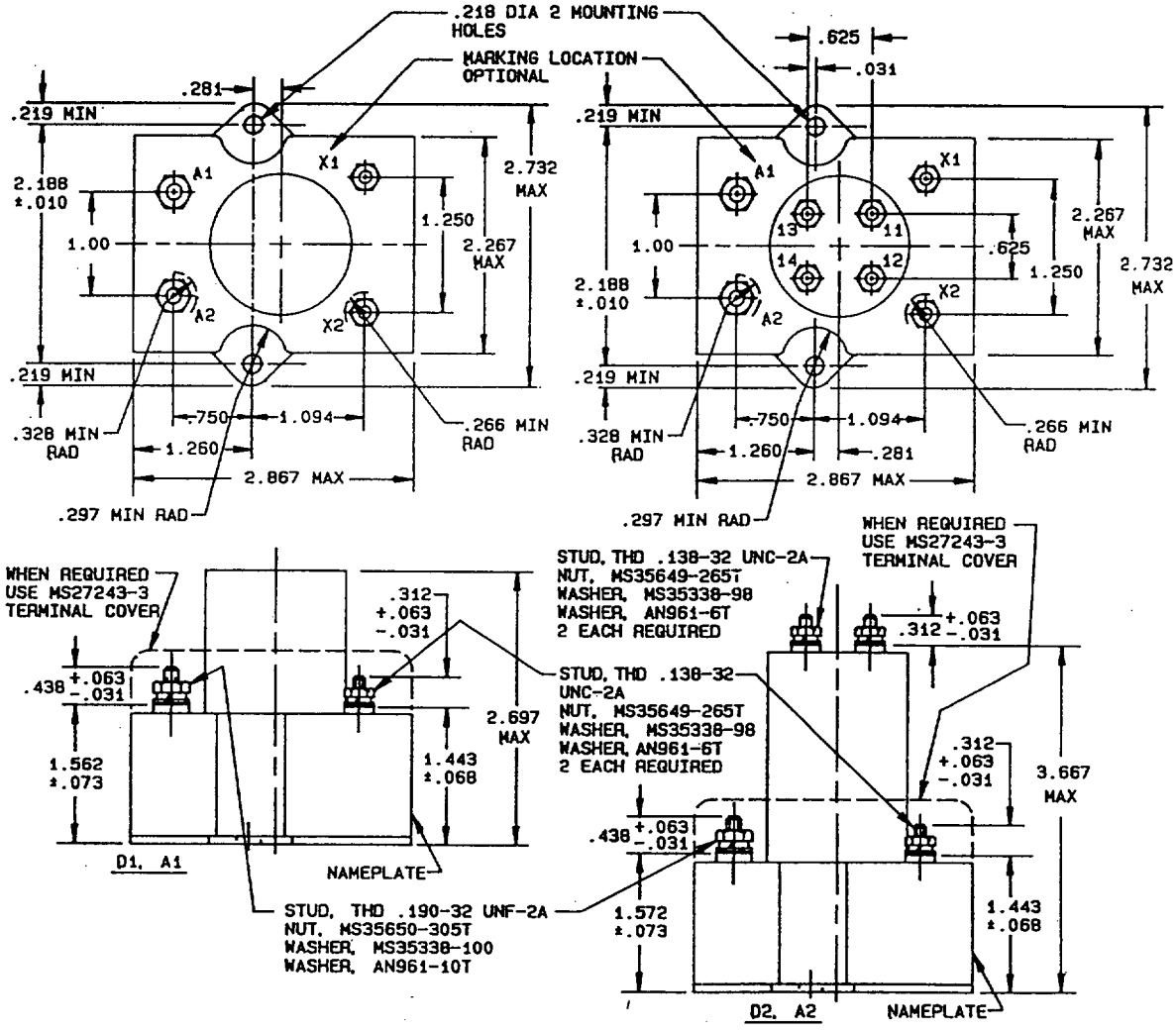
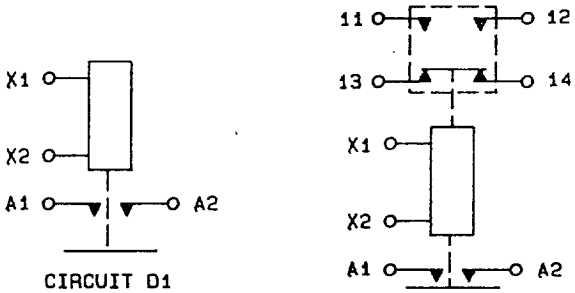


FIGURE 1. Dimensions and configurations.

(L) denotes changes

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CIRCUIT DIAGRAMS

Inches	mm	Inches	mm
.010	0.25	.625	15.86
.031	0.79	.750	19.05
.063	1.60	1.000	25.40
.068	1.73	1.094	27.79
.073	1.85	1.250	31.75
.138	3.51	1.260	32.00
.190	4.83	1.443	36.65
.218	5.54	1.562	39.67
.219	6.96	1.572	39.93
.240	6.10	2.188	55.58
.266	6.76	2.267	57.68
.281	7.14	2.697	68.50
.297	7.54	2.732	69.39
.312	7.92	2.867	72.82
.328	8.33	3.667	93.14
.438	11.13		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031.
4. This specification sheet takes precedence over documents referenced herein.
5. Referenced documents shall be of the issue in effect on date of invitation for bid.
6. Coil and auxiliary terminals may use additional flat washer for terminal seat.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Dash numbers and general characteristics: See table I.

Contact data:

Load ratings: see table II.

Maximum contact drop:

Initial: 0.150 volt.

After life test: 0.175 volt.

Overload current (NO): 400 amperes.

Rupture current (NO): 500 amperes.

Coil data: See table III.

Duty rating: Continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac-operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength (sea level): 2-5 seconds (see table IV).

Dielectric strength (altitude): 1 minute (see table V).

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock G-level: 25 g's.

Duration: 6-9 ms.

Maximum duration contact opening: 2 ms.

Vibration (sinusoidal): See table VI.

Accelleration: 15 g's.

Qualification by similarity: See table VII.

Part or Identifying Number (PIN): MS24140 - (plus dash number from table I).

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TABLE I. Dash numbers and general characteristics. 1/

PIN MS24140-	Type	Coil	Terminal type	Mounting or mating socket	Auxiliary contacts	Maximum weight in pounds 2/
D1	I	dc	Stud	Flange	None	.95
D2					Yes	1.10

1/ Dash numbers A1 and A2 have been cancelled without replacement.
2/ Weights include covers and barriers.

(L) TABLE II. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	50	50		5	5	50		5					
Inductive	10	50		5	5	50		5					
Inductive													
Motor	50	50				50							
Lamp	50	25		.75	.75	10 2/		.75					
Transfer load													
Mechanical life reduced current	100	12.5		1.25	1.25	1.25		1.25					
Intermediate current	50	5		Per	spec	5							

1/ Absence of value indicates relay is not rated for 3 phase application.
2/ Test requirement: 100 amperes inrush current with 2 seconds off-time.

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(L) TABLE III. Operating characteristics.

PIN MS24140-	Coil data											Time - milliseconds max					
	Coil	Rated			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Bounce			
		Volts 1/	Freq Hz	Ω Res +15% -10	Volts	Amperes	Normal 2/	High temp test	Cont current test					Main		Aux	
														NO	NC	NO	NC
D1	X1,X2	28	dc	78	29	.50	18	21	22.5	7	1.5	20	10	2	---	---	---
D2	X1,X2	28	dc	78	29	.50	18	21	22.5	7	1.5	20	10	2	---	4	4

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.

TABLE IV. Dielectric strength (sea level).

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250 V rms	1,500	1,000 V rms	1,125
Auxiliary contacts	1,250 V rms	1,500	1,000 V rms	1,125
All other points	1,250 V rms	1,500	1,000 V rms	1,125

TABLE V. Dielectric strength (altitude).

	28 V dc	115 V ac
Coil to case	500 V rms	500
Auxiliary contacts	500 V rms	500
All other points	500 V rms	500

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TABLE VI. Vibration levels (sinusoidal).

PIN MS24140-	5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1,500 Hz
D1	.08 DA	.06 DA	10 g's	10 g's	8 g's
D2	.08 DA	.06 DA	10 g's	6 g's	6 g's

TABLE VII. Qualification by similarity.

PIN MS24140-	Loads						Dynamics 1/ 3/			Environmental		
	Type 1			Type 1 ER								
	A	B	C	D	E	F	A	B	C	A	B	C
D1	4						2			4		
D2	4 2/						2			4 2/		

1/ For group C testing, each dash number shall be tested to meet vibration requirements.
2/ For testing of relays with auxiliary contacts see MIL-R-6106, Appendix I.
3/ All relays must be tested. Reference MIL-R-6106, Appendix I.

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 85

Review activities:
Navy - EC
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0873)

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